

ORIGINAL ARTICLE

YOLO-based panoramic dental X-ray image analysis

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Abstract

Despite the considerable progress in dentistry, the automated detection and diagnosis of oral pathologies remains a major challenge. The great variability of the lesions, the subtleties of some of them, and the anatomical variations greatly complicate this task, even for high-performance AI systems. This article presents a three-step pipeline exploiting deep learning techniques to meet these challenges and enable automated detection of dental diseases from panoramic X-rays. The proposed pipeline uses different YOLO models (v5, v6, v7, v8, and v9) with variable sizes (nano, small, medium, large, and x-large) in order to segment the dental quadrants, accurately detect each individual tooth, and finally detect the main pathologies, such as caries or impacted teeth. The results obtained demonstrate the high potential of these YOLO models to overcome the limitations of previous approaches, thus paving the way for more reliable and efficient automated dental analysis solutions.

Keywords Panoramix X-ray · Disease detection · Dentistry · Deep learning · Segmentation

1 Introduction

X-ray imaging plays an essential role in dental care, enabling health professionals to examine the structure of teeth, gums, and alveolar bones in detail [1, 2]. Among the most commonly used imaging techniques, panoramic X-rays stand out for their ability to capture an overview of the upper and lower jaws in a single image [3].

Panoramic X-ray is a dental imaging technique with multiple clinical applications [4]. It enables analysis of the morphology of facial and dental structures, including the maxillary and mandibular arches, as well as supporting tissues [5, 6]. It is used to detect pathologies in the dentomaxillofacial region [7–9], assess tooth growth and development [10], identify anomalies in dental and maxillofacial structures [11], and estimate a patient's age based on radiographic features [12, 13].

In addition, panoramic X-rays are invaluable for identifying jaw fractures [14, 15], locating wisdom teeth [16], diagnosing dental and bone diseases [17], detecting root remnants [10], assessing impacted teeth [18, 19], diagnosing temporomandibular joint dislocations, as well as dental anomalies and surrounding tissues [20, 21]. This imaging examination provides a comprehensive view of facial structures, making it an essential tool for diagnosis, treatment planning, and patient follow-up in dentistry [22, 23].

However, interpreting these panoramic images remains a complex task, requiring in-depth expertise on the part of dentists and radiologists [24, 25]. Traditional image analysis models can sometimes prove limited, particularly when it comes to locating and numbering teeth and detecting pathological anomalies.

In this context, artificial intelligence (AI), specifically deep learning (DL), has emerged as a promising approach for the automated examination and analysis of dental panoramic X-ray images [26]. However, the development of automated algorithms to analyze panoramic radiographs remains complex due to anatomical variability [27] and the lack of accessible annotated data [28]. Despite these difficulties, it would be wrong to



overlook the potential benefits of integrating AI into dental radiology analysis [29]. Indeed, real progress could be made in terms of clinical outcomes and patient satisfaction [30].

The use of DL to analyze panoramic dental X-ray images represents a major advance for dentistry [31–33]. By automating certain examination and diagnostic tasks, this technology has the potential to improve the accuracy, efficiency, and accessibility of dental care, to the direct benefit of patients.

Indeed, DL enables faster, more accurate image analysis, reducing the time dentists have to spend interpreting results. This frees up valuable medical time that can be dedicated to essential clinical tasks, improving the overall efficiency of dental care.

In addition, DL techniques can potentially contribute to reducing diagnostic errors. In fact, DL's algorithms have demonstrated their remarkable effectiveness in various tasks related to the analysis of panoramic radiographs, in particular the automatic segmentation of anatomical structures, the automatic numbering of teeth, and the precise detection of anomalies or early signs of dental pathology, thus surpassing traditional methods. This ability to identify problems at an early stage allows for a faster and more appropriate intervention, thus avoiding potential complications for patients.

Beyond these immediate benefits, the integration of DL into panoramic X-ray analysis also opens up new research perspectives. AI models can be used to better understand the links between radiographic features and dental pathologies, contributing to the continuous improvement of disease screening and monitoring practices.

The main contributions of this paper are:

- Recent YOLO models, such as YOLOv9, are optimized for the precise detection and location of infected teeth, as well as the identification of the quadrants to which they belong. These models make it possible to considerably reduce false detections, while significantly improving the performance of deep learning-based methods for the detection and segmentation of teeth and dental quadrants.
- YOLO models, like YOLOv9e, have indeed demonstrated good performance for the precise segmentation of dental quadrants on X-ray images. This ability to precisely localize different regions of the mouth is essential for a detailed analysis of oral health status.
- The YOLO models have also shown great efficiency in the detection and enumeration of individual teeth, as well as in the identification of the quadrant to which each tooth belongs. This ability to provide key information on the location of the teeth is of crucial importance for the monitoring and diagnosis of dental pathologies, given the anatomical complexity of the oral cavity and the inter-individual variability that results from it.
- Although the YOLO models still encounter certain limitations in the accurate detection of dental pathologies by identifying the tooth number and the quadrant involved, they have demonstrated a strong potential in this field.

The rest of this article is structured as follows: Section 2 reviews previous research on the use of deep learning techniques for the detection and segmentation of dental quadrants, the precise enumeration of teeth, as well as the detection of dental diseases. Section 3 presents in detail the datasets used in the framework of this study, the most recent YOLO models that have been used, as well as the different evaluation measures chosen to analyze performance. In Section 4, the results of the tests and experiments carried out are discussed and analyzed in depth. Finally, Section 5 concludes the work.

2 Background

In dentistry, the analysis of X-ray images relies mainly on the experience and visual perception of dentists. However, the rotation of the X-ray tube around the patient poses problems for obtaining panoramic dental images. The efficiency of the examination can be affected by several problems caused by this acquisition method. These include machine noise, a lack of contrast in contours, and overlapping anatomical structures.

These factors can make image analysis less precise and more difficult for dentists. Machine-generated noise can compromise the clarity of detail, while low-contrast contours make it more difficult to distinguish between different areas. Additionally, overlapping anatomical structures can make it challenging to identify and locate dental features accurately.

These limitations in the visual analysis of panoramic X-rays underline the importance of introducing automated approaches based on ML [34]. Specific models have been developed for precise tasks such as segmentation of dental quadrants [35, 36], individual tooth detection [37, 38], enumeration of teeth in quadrants [39–41], automated diagnosis of some anomalies [42, 43], and the development of appropriate treatment plans [27, 44]. Each of these models was designed to solve a specific problem, such as quadrant segmentation, tooth localization, or the detection of certain conditions.

Instead of examining each component separately, it might be more efficient to take an integrated approach that incorporates these multiple tasks into a single model. Such an approach would produce more useful results by more effectively accounting for the links and interactions between the various factors under analysis.

It is necessary to create automated methods that can recognize and categorize dental diseases on radiographs, as well as segment and number teeth. There are several obstacles in the way of automatically analyzing dental X-ray pictures. Among these is the accurate segmentation of the dental anatomy to identify and differentiate each anatomical component (bone, soft tissue, teeth, etc.) on the images.

An in-depth literature review of traditional tooth segmentation techniques on dental panoramic X-rays is presented by Silva et al. [45]. In addition, they present an extensive dataset called the UFBA-UESC dental image dataset [45, 46]. The authors concluded from experiments conducted on this dataset that the Mask R-CNN solution [47], which considers the whole tooth arch as a single instance, is significantly better than traditional techniques. The first study to show the benefits of using deep learning in panoramic radiographs was this one.

The proposed method by Xia et al. [48] included the upper and lower mandibles of natural contact scanning computed tomography images of the upper and lower teeth, allowing for the segmentation of a single tooth from those pictures. This produced a single completed dental model. Koch et al. [49] used the U-Net [50], on the UFBA-UESC dataset to perform semantic segmentation on panoramic dental images, resulting in improved sample segmentation and a smaller, simpler network design.

Kong et al. [51] have developed an Efficient Encoder-Decoder Network (EED-Net) architecture to segment the maxillofacial quickly and accurately. The network is composed of a residual encoder, a multi-feature extractor, and an object-oriented decoder. Zhao et al. [35] introduced TSASNet, a two-stage network, to segment teeth in dental panoramic X-ray images. First, the tooth region was typically located using an attention model, and then, in a second phase, a fully convolutional network was utilized for accurate segmentation. Using a regional mask neural convolution network (Mask R-CNN), Tekin et al. [52] segmented and numbered teeth on occlusal radiographic images. They produced high-quality segmentation masks.

In their paper, Sheng et al. [53] employed the SWin-Unet architecture, a U-shaped encoder-decoder model based on transformers with skipped connections, to perform segmentation of panoramic radiographs. To thoroughly evaluate the performance of dental segmentation with SWin-Unet, the researchers introduced the PLAGH-BH dataset for research purposes [53]. Compared with reference models such as U-Net, Link-Net, and FPN, SWin-Unet demonstrated better performance in the dental segmentation task on the PLAGH-BH dataset [53].

Kanwal et al. [54] presented an approach based on panoptic segmentation that combines instance segmentation with semantic background segmentation. They proposed an architecture specifically designed for dental segmentation, using a network based on a dual-path transformer. This network is associated with a quality panoptic loss function that improves the model training process. The model is able to directly predict tooth masks and their corresponding classes. Using the UFBA-UESC dental image dataset, the researchers observed a significant improvement in segmentation performance. This approach outperformed existing methods in terms of segmentation efficiency and accuracy.

Giulia et al. [55] have also used the Mask R-CNN model for tooth segmentation in panoramic dental radiographs. This model combines both bounding box object detection and semantic instance segmentation, enabling

the precise identification and delineation of individual tooth contours in radiographic images. The model was trained and validated using the Tufts Dental Database. The results demonstrated high performance in terms of detection accuracy and Dice index, confirming its effectiveness for automated tooth segmentation on dental panoramic radiographs.

Senbao et al. [56] proposed a Teeth U-Net model to address the problem of accurate segmentation of dental teeth. This model relies on a dense connection mechanism between the encoder and decoder, intending to resolve the problem of blurred contours at tooth boundaries.

Sun et al. [57] developed a model called STS-TransUNet, which is a hybrid architecture that combines the strengths of both CNNs and Transformer models. The model employs a U-shape architecture and integrates a hybrid encoder that merges the strengths of CNN and Transformer to enhance positional awareness and long-range information fusion. Additionally, the model incorporates the Convolutional Block Attention Module (CBAM) to improve spatial and channel information management, contributing to exceptional performance. The architecture also includes a self-training strategy with pseudo-labels, data augmentation, and iterative model optimization for semi-supervised training, effectively improving performance with limited labeled data.

Erin et al. [58] have proposed a collaborative learning model that combines two distinct deep learning models for tooth segmentation and filling segmentation. The tooth segmentation model is designed to differentiate between primary and permanent teeth in mixed dentition scenarios, while the filling segmentation model focuses on identifying different types of dental fillings. The collaborative model integrates the results of both models to provide a more accurate and comprehensive analysis of dental radiographs. The design of the collaborative model enables it to conclude multiple models using a single input for a variety of relevant tasks, in this case, tooth segmentation and filling segmentation. Model accuracy is determined as the weighted average of the accuracy of the individual models, and the integrated inference summary is saved in a standard format (Microsoft COCO) that can be used for further analysis.

Several studies have focused on the automatic detection and numbering of teeth on panoramic radiographs. In previous studies, the identification of teeth from dental images has been the focus of attention. Oktay et al. [59] used classical artificial neural networks to recognize the type of each tooth after localizing the oral cavity. However, no numbering system was employed. Later, Tuzoff et al. [40] improved the method by also enabling teeth in panoramic radiographs to be numbered using the Faster R-CNN architecture [60]. Due to the use of several deep learning algorithms at different stages, their method has a high computational cost.

Chen et al. [61] developed a model based on the Faster R-CNN architecture to detect and number teeth in dental periapical films. The authors presented a simple strategy known as “three post-processing techniques” to improve the outcomes of Faster R-CNN. Because the approach utilized is effective for periapical coatings, it is definitely not appropriate for other teeth.

In their research on panoramic dental radiographs, Kim et al. [62] identified periodontal bone loss by selecting only the tooth numbers corresponding to the lesion rather than all the teeth in the image. An original approach was proposed by Kim et al. [63], combining a regional convolutional neural network (R-CNN) for instance detection, a single-shot multi-box detector (SSD), and heuristic methods to number teeth in a panoramic X-ray image. Although this combination of techniques improved performance over the basic algorithm, several quantitative evaluation indices remained vulnerable to improvement, according to the authors.

Mahdi et al. [64] developed an automated tooth detection approach that enhanced their findings utilizing the Faster R-CNN technology, as well as a candidate optimization strategy that considers both the positional relationship and the confidence score of the candidates. Using the recommended optimization strategy, they could enhance the model results even further.

Tekin et al. [65] proposed a model based on the Mask R-CNN, a deep learning model that extends Faster R-CNN by adding a branch for predicting segmentation masks on each Region of Interest (RoI), identification, and numbering after performing tooth labeling using the software designed by DentiAssist.

Ahmet et al. [41] proposed a new approach based on the R-CNN mask model and a heuristic algorithm. The approach consists of three main modules: tooth numbering, overlap box removal, and incorrect numbering

correction. The tooth-numbering module uses the R-CNN mask model. The R-CNN mask model is used, for instance, for segmentation, which involves the detection and identification of each object of interest in an image. In addition, a heuristic algorithm is proposed to improve the performance of the tooth numbering method based on the basic deep learning model. The combination of the deep learning model and the heuristic algorithm enables accurate tooth numbering in panoramic images, demonstrating high accuracy in tooth numbering.

3 Materials and methods

3.1 Dataset

The DENTEX dataset [66] consists of panoramic dental X-rays collected from three different institutions, taken under standard clinical conditions but with varying imaging protocols and equipment. This diversity in image quality reflects the heterogeneity of practices. The sample consists of radiographs from Patients aged at least 12 years, randomly selected from hospital databases to preserve anonymity and confidentiality.

As shown in Fig. 1, the DENTEX dataset provides three hierarchically annotated sets to facilitate various dental detection tasks:

1. *Quadrant detection* 693 annotated images solely for quadrant location. Although this task may seem redundant, it is essential to enable the use of the “Fédération Dentaire Internationale” FDI numbering system.
2. *Tooth enumeration* 634 annotated images for individual detection of teeth in their respective quadrants.
3. *Anomaly detection* 1005 annotated images for precise localization of aberrant teeth via enumeration and diagnosis.

The FDI dental numbering system is a standardized nomenclature used internationally. As illustrated in Fig. 2, it assigns a unique numerical code to each anatomical element of the mouth, enabling precise identification.

Classification begins by dividing the oral cavity into 4 quadrants numbered 1 to 4. The upper right quadrant is assigned the number 1, the upper left 2, the lower left 3, and the lower right 4. Each tooth is then individually coded from 1 to 8 within its respective quadrant. The number 1 is allocated to the upper central tooth, with numbers increasing toward the back of the jaw. For example, the third lower left tooth, or wisdom tooth, is assigned the number 48. This double number means that it is tooth no. 8, located in the lower left quadrant no. 4.

Thanks to its simple, standardized coding system, the FDI classification provides a common language to facilitate exchanges between oral health professionals worldwide.

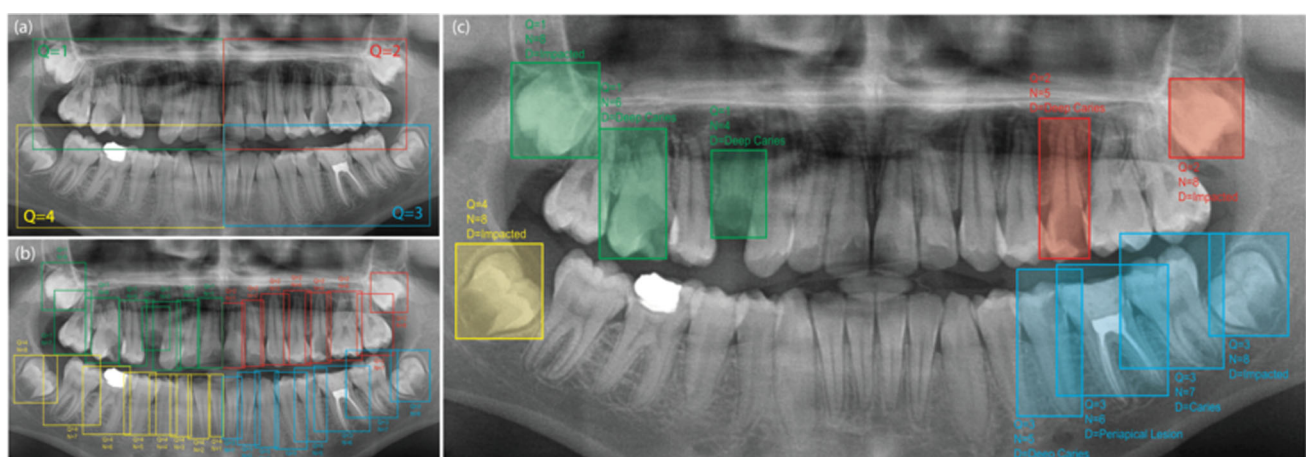
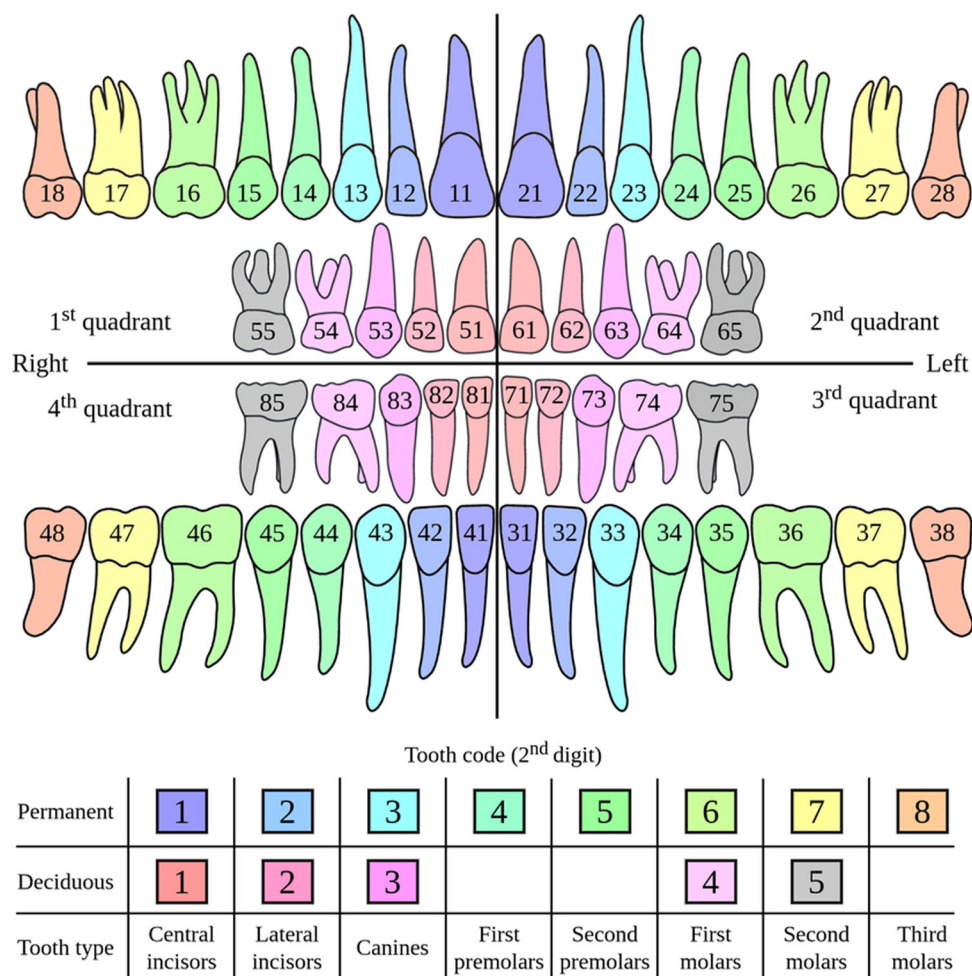


Fig. 1 **a** quadrant-only for quadrant detection, **b** quadrant-enumeration for tooth detection, and **c** quadrant-enumeration-diagnosis for abnormal tooth detection [66].

Fig. 2 FDI World Dental Federation notation system: Each tooth is represented by a two-digit number. The first digit denotes the quadrant of the tooth and whether it is permanent or deciduous, while the second digit indicates the type of tooth. Color codes are also used to convey qualitative results [67].



Quadrant segmentation set 1 greatly facilitates dental enumeration task 2 by first locating each tooth in its quadrant. Although only Level 3 data are subject to formal evaluation, Intermediate Sets 1 and 2 enable models to be progressively learned on tasks of increasing complexity.

The DENTEX dataset benefits from a rigorous annotation process to ensure the quality and reliability of the metadata. Each image is individually annotated by an advanced dental student. These primary annotations are then validated and corrected by one of the three expert dentists associated with the project. These experts each have more than 15 years of clinical experience, making them high-level referees for the critical review of student annotations. Thanks to this two-stage annotation methodology, which combines the students' theoretical knowledge with confirmed practical expertise, the data in the DENTEX set benefits from an exceptional level of quality and accuracy.

3.2 Proposed approach

In our study, we adopted a 3-step pipeline to detect disease, tooth number, and corresponding quadrant from dental panoramic X-rays, as shown in Fig. 3. Step 1 consists of quadrant segmentation by applying segmentation models such as YOLOv9-Seg specifically trained to isolate each of the 4 quadrants of the upper and lower jaw in specific areas of the image. Step 2 performs individual tooth detection using YOLO-seg models trained to precisely locate each tooth, associating it with a number between 1 and 32, and identifying the quadrant among upper right/left or lower right/left. Finally, step 3 performs disease detection on each tooth by training one of the YOLO models, trained on patches centered around each tooth from step 2, to predict diseases such as caries, deep

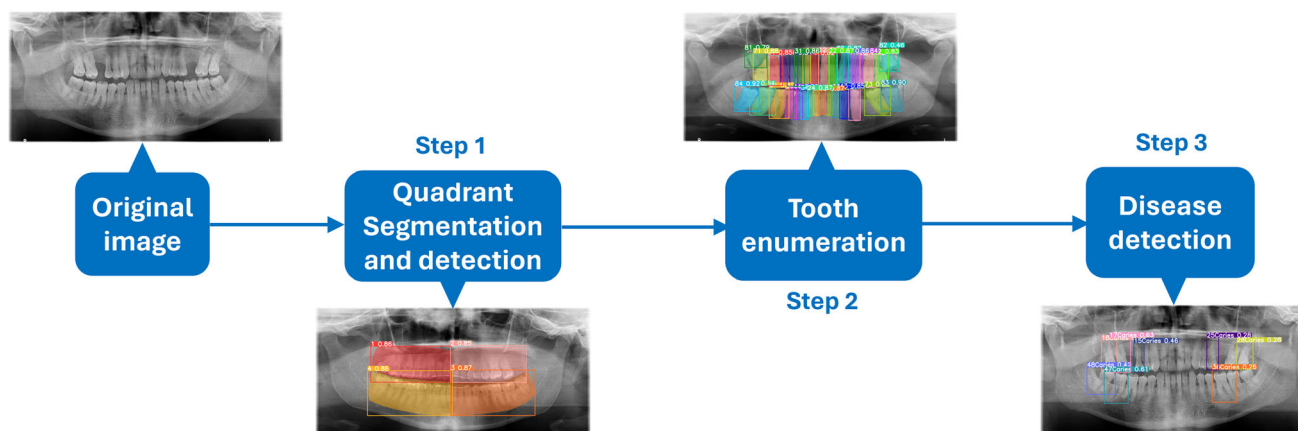


Fig. 3 Proposed pipeline for dental disease detection.

caries, periapical lesions, and impacted teeth. By combining the results of the 3 steps, we obtain for each tooth its location, its number, the quadrant, and the disease detected, thus constituting a complete automated diagnostic pipeline.

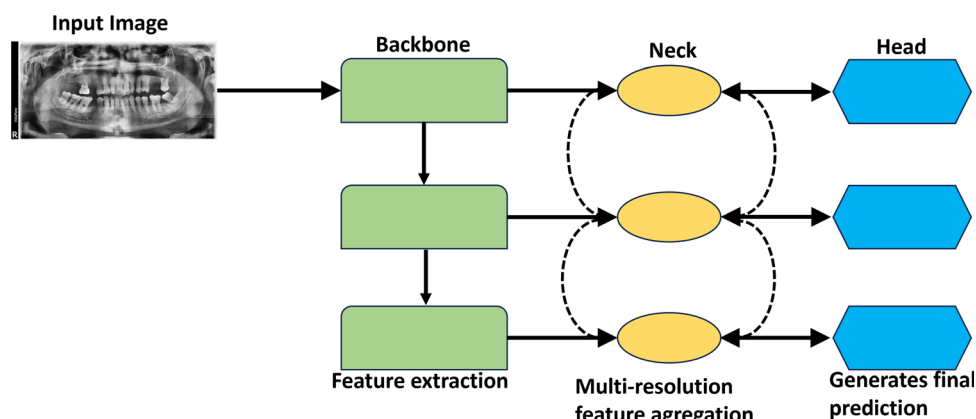
To implement these architectures, we used open-source code available on the Ultralytics repository. We also opted for transfer learning, initializing the models with the weights of a network pre-trained on COCO. This transfer learning step aimed to accelerate training on our dental data, by taking advantage of the knowledge previously acquired on other objects. Therefore, the recognized accuracy and speed of YOLO models, combined with initialization by transfer learning, proved the best approach for our tasks.

3.3 YOLO Model

In recent years, the YOLO (You Only Look Once) series of methods [68–70] have become increasingly popular for real-time object detection. They have been widely adopted in various application scenarios requiring high speed. While the YOLO algorithm enables efficient and accurate detection of objects in an image, the concept of instance segmentation goes a step further by performing pixel-by-pixel segmentation of regions identified through object detection. This process allows for a more detailed understanding of the image content. In the case of YOLO, it combines object detection with semantic segmentation techniques. By combining all bounding boxes classified as the same object at a pixel level, it is possible to obtain accurate segmentation results for object entities.

The architecture of YOLO object detectors consists of three main components, as shown in Fig. 4:

Fig. 4 YOLO models' Architecture.



1. The backbone plays an essential role in extracting valuable features from input images. It typically uses a convolutional neural network (CNN) pre-trained for large-scale image classification tasks, such as ImageNet. The backbone maps hierarchical features at various scales by extracting lower-level features, such as edges and textures, from previous layers. Higher-level features, such as object parts and semantic information, are then deleted in deeper layers.
2. The neck's role is to fine-tune the features extracted by the backbone by merging the information. It is used to combine high-level and low-level feature maps. It usually consists of an FPN (Feature Pyramid Network) [71, 72] module or an enhanced version such as PANet (Path Aggregation Network) [73] or Bi-FPN (Bidirectional Feature Pyramid Network) [74]. The FPN module subsamples low-level maps and merges them with high-level maps, thus enhancing their spatial information. Enhanced modules such as PANet and Bi-FPN add a step where smaller maps are upsampled and merged with larger ones, enhancing their semantic information. By combining low- and high-level information via subsampling and oversampling operations, the neck plays a role in fine-tuning the features between the backbone and the detection head.
3. The head exploits the features refined by the neck to simultaneously predict bounding boxes and associated classes. It is composed of convolutional or fully connected, layers. The head can take two forms: coupled, where a single structure both predicts boxes and performs classification, or decoupled, with one branch dedicated to localization and another to classification. Research [75] has shown that the decoupled structure offers better performance in terms of accuracy and convergence during training.

This three-block architecture (backbone, neck, head) enables algorithms like YOLO to perform object detection efficiently, extracting and then progressively refining visual information through a succession of multi-scale convolution and fusion operations. Head decoupling also enhances performance in terms of precision and convergence.

3.3.1 YOLOv5

YOLOv5, developed by Ultralytics, incorporates the AutoAnchor algorithm, which automatically checks and adjusts anchor boxes according to data and training parameters. This optimizes detection accuracy by adapting anchor boxes to the image size. The AutoAnchor algorithm uses a combination of k-means and genetic evolution algorithms to obtain accurate initial conditions.

Its architecture is built around a backbone composed of a CSPDarknet53 [76] modified at its origin by a large-window-stridden convolution layer that reduces computational costs. This backbone extracts basic visual features via successive convolution operations. The neck takes over, refining these features via the Spatial Pyramid Pooling Fast (SPPF) [77] module, which processes information at different scales in an accelerated way, and the modified CSP-PAN module. It also consolidates feature maps through operations on their resolution. The head then processes this refined information. It contains three separate but simultaneously trained parts for bounding box prediction, classification, and instance mask generation. Its structure is similar to that of YOLOv3.

Five versions are available, depending on the accuracy and speed required, from the nano model to the extra-large version. The PyTorch/CUDA framework is used for training via techniques such as Adam or learning rate schedulers.

3.3.2 YOLOv6

Meituan Vision's AI department developed YOLOv6 [78]. YOLOv6 introduced several major improvements over its predecessors. Firstly, its EfficientRep [79] backbone makes more efficient use of the computing power of hardware such as the GPU. Indeed, this backbone boasts strong representation capabilities while optimizing resource use. Secondly, in terms of neck, the Rep-PAN network has been designed based on a hardware-sensitive structure to achieve both a more efficient use of resources and a better balance between precision and speed.

Finally, YOLOv6 adopts a decoupled head whose design takes into account the balance between operator representation capacity and induced computational load, thus simplifying the overall architecture.

3.3.3 YOLOv7

YOLOv7 [70] has introduced architectural modifications and a series of “bag-of-freebies” to increase accuracy without impacting inference speed, only training time. Firstly, new E-ELAN [80] blocks have been introduced in the backbone. These blocks combine features from different groups using a shuffle and cardinality merge mechanism. This improves model learning. In addition, the RepVGG [81] module used previously has been modified to remove identity connections.

3.3.4 YOLOv8

YOLOv8 [82] has made architectural improvements while retaining high performance and versatility for different tasks. It is based on a backbone similar to YOLOv5 but with modifications to the C2f module (cross-stage partial bottleneck with 2 convolutions). This C2f module combines high-level features with contextual information to improve detection accuracy. YOLOv8 is also an anchorless model, with a decoupled head handling objectness, classification, and regression tasks independently, allowing each branch to concentrate on its task and improving overall accuracy. The loss functions used are (Complete Intersection over Union) CIoU [83] and (Distribution Focal Loss) DFL [84] for bounding boxes, and a binary cross-entropy for classification. At the output layer level, YOLOv8 exploits the sigmoid as an activation for the objectness score, representing the probability that a box contains an object, and the softmax for class probabilities, representing a member of each possible class.

3.3.5 YOLOv8-seg

YOLOv8-seg [85] is an extension of the YOLOv8 model that incorporates both object detection and semantic segmentation tasks. While the original YOLOv8 model focuses on object detection, YOLOv8-seg combines the capabilities of detecting objects and performing pixel-wise semantic segmentation in a single model.

In traditional object detection, bounding boxes are used to localize and classify objects in an image. On the other hand, semantic segmentation assigns a class label to each pixel in an image, effectively segmenting the image into different regions corresponding to different object classes.

YOLOv8-seg achieves this by modifying the YOLOv8 architecture to incorporate an additional branch for semantic segmentation. This allows the model to simultaneously predict object-bounding boxes and perform pixel-level segmentation within the same network. By leveraging shared feature representations, YOLOv8-seg can effectively handle both tasks in a unified manner.

By combining object detection and semantic segmentation, YOLOv8-seg provides a comprehensive understanding of the visual scene, enabling more detailed analysis and interpretation of images. This makes it particularly useful in applications where both object localization and pixel-level segmentation are required, such as autonomous driving, medical imaging, and scene understanding.

3.3.6 YOLOv9

YOLOv9 is the latest version of the YOLO object detection model developed by Wang et al. [86]. It follows on from YOLOv7, which brought significant improvements, notably thanks to the training of gift bags, enabling detection accuracy to be increased without impacting real-time performance. However, YOLOv7 did not resolve the problem of information loss during the propagation of input data through the network, commonly known as the information bottleneck. This challenge results from downscaling operations that can dilute crucial data. While solutions such as reversible architectures, masked modeling, and deep supervision limit this phenomenon, they

also have their drawbacks. What's more, they are less effective on the compact models required for real-time detection.

YOLOv9 proposes two key techniques to overcome the limitations of previous approaches: programmable gradient information (PGI) and generalized efficient layer aggregation network (GELAN) [86–88]. The use of PGI can provide complete input information for the target task, enabling the objective function to be calculated accurately and reliable gradient information to be obtained for updating the network weights. In addition, a new lightweight network architecture, called GELAN based on gradient planning, has been developed. The results obtained confirm the superior performance of PGI on lightweight models using the GELAN architecture.

These methods directly solve the informational bottleneck problem, improving the accuracy and efficiency of object detection.

3.4 Evaluation metrics

To evaluate performance and compare the different models in the YOLO family, precision and recall were measured using confusion matrices. These matrices are based on 4 key parameters derived from a binary classification: true positives (T_p) (correct detection of the target), false positives (F_p) (erroneous prediction of the target), true negatives (T_n) (correct identification of an image not containing the target), and false negatives (F_n) (non-detection of the real target).

In fact, a true positive corresponds to the case where the YOLO network correctly identifies the quadrant or tooth sought. A false positive refers to an inaccurate prediction of the target. A true negative occurs when the network correctly recognizes an image that does not contain the target. Finally, a false negative results from a failure to predict the actual target. Analysis of these 4 types of events enables us to assess the performance of YOLO models in their ability to detect and classify correctly or not the targets studied.

- **Precision (P)** (1) is a measure that quantifies the proportion of true positives among all the positive predictions made by the model. It thus assesses the model's ability to limit the number of false positives, i.e. to avoid falsely predicting the presence of the desired target:

$$P = \frac{T_p}{T_p + F_p} \quad (1)$$

- **Recall (R)** (2) is a measure used to evaluate a model's ability to detect all positive examples. It measures the proportion of positive examples correctly identified as such by the model, relative to the total number of positive examples:

$$R = \frac{T_p}{T_p + F_n} \quad (2)$$

- **The Intersection over Union (IoU)** (3) measure was used during training to assess the overlap between predicted bounding boxes and annotation-based boxes. The IoU rate was used to determine whether a detected object, such as a tooth disease, should be considered a true positive. Different IoU thresholds ranging from 0.5 to 0.95 were applied for comparison purposes. It is calculated as:

$$\text{IoU} = \frac{\text{Area of intersection}}{\text{Area of Union}} = \frac{|B_p \cap B_g|}{|B_p \cup B_g|} \quad (3)$$

where:

- B_p : Predicted bounding box.

- B_g : Ground truth bounding box.
 - $|B_p \cap B_g|$: Area of intersection.
 - $|B_p \cup B_g|$: Area of union, computed as $|B_p| + |B_g| - |B_p \cap B_g|$.
 - $\text{IoU} = 1$: Perfect alignment.
 - $\text{IoU} = 0$: No overlap.
 - A threshold (e.g., 0.5) is often used to classify a prediction as correct (True Positive).
- **Average Precision (AP)** (4) corresponds to the average of the precision values obtained for different recall levels, varying between 0 and 1. It can be seen as the area under the Precision-Recall Curve (PR Curve), which is plotted by varying the confidence threshold of the model predictions. The PA measures the area under the PR Curve by interpolating its component points.

$$\text{AP} = \int_{r=0}^1 p(r) dr \quad (4)$$

Where:

- $p(r)$ is the precision for a given recall r .
 - The integral is calculated for all recall values r between 0 and 1.
- **The mean average precision (mAP)** (5) was also used to assess the performance and detection capability of the YOLO models. It is calculated as the average AP (Average Precision) of each class, where AP summarizes the model's recall and precision in detecting a specific class. Unlike AP, mAP aggregates the results of all classes to give an overall picture of the model's capabilities. A high mAP value, close to 1, indicates that the model correctly identifies the presence and boundaries of different objects in all classes, reflecting good overall detection performance. It is calculated as:

$$\text{mAP} = \frac{1}{N} \sum_{n=1}^N \text{AP}_{\text{class}(n)} \quad (5)$$

Where:

- AP_{class} is the AP calculated for a given class.
- N is the total number of classes.

Two variants of mAP were calculated: mAP50 (6) for an IoU threshold of 0.5, and mAP50-95 (8) by varying this threshold from 0.5 to 0.95. These measurements provide insight into the accuracy with which YOLO models can correctly identify different bee species. The IoU was used to classify a detection as true or false positive, based on the overlap of boxes. The mAP provides information on the overall accuracy of predictions for different levels of spatial requirement, summarizing the performance of the models.

- **mAP50 (Mean Average Precision at IoU=0.5):**

$$\text{mAP50} = \frac{1}{N} \sum_{i=1}^N \text{AP}_{\text{class}_i} \quad (6)$$

where:

- N : Number of classes.
- $\text{AP}_{\text{class}_i}$: AP for class i at $\text{IoU}=0.5$.

mAP50 evaluates model performance with a relatively lenient localization requirement ($\text{IoU} \geq 0.5$).

- **mAP50:95 (Mean Average Precision from IoU=0.5 to 0.95)** is the mean AP calculated across IoU thresholds from 0.5 to 0.95 (in steps of 0.05), averaged over all classes. It assesses performance across a range of localization precisions, making it a stricter metric.

1. Compute AP for each IoU threshold $t \in \{0.5, 0.55, 0.6, \dots, 0.95\}$.
2. Average AP across IoU thresholds:

$$AP_{50:95} = \frac{1}{10} \sum_{t=0.5}^{0.95} AP_{\text{class}_t} \quad (7)$$

3. mAP50:95:

$$\text{mAP}_{50:95} = \frac{1}{N} \sum_{i=1}^N AP_{50:95, \text{class}_i} \quad (8)$$

where:

- N : Number of classes.
- $AP_{50:95, \text{class}_i}$: Average AP across IoU thresholds for class i .

mAP50:95 is more stringent, as it includes high IoU thresholds (up to 0.95), requiring precise localization.

4 Results and discussion

For each step in our pipeline, we divided the corresponding dataset into 3 distinct subsets:

- 80% of the images were assigned to the training set, on which the model would learn the image features.
- 10% of the images constituted the validation set, on which the model would be tested and adjusted as it learned.
- The remaining 10% made up the test set, designed to evaluate the model's final performance once training was complete.

4.1 Quadrant segmentation

Quadrant segmentation in dental radiography is a crucial task for the accurate analysis of dental panoramic images, enabling different anatomical and pathological regions to be correctly identified. To this end, our dataset, specifically dedicated to quadrant processing, is divided into three subsets: training, validation, and testing.

Table 1 evaluates the quadrant segmentation performance obtained with different Yolo-seg models, compared with the results published by Hamamci et al. [89] and He et al. [90].

The results presented in Table 1 indicate the very good performance of the different versions of the YOLO-seg model, including YOLOv5-seg, YOLOv7-seg, YOLOv8-seg, and YOLOv9-seg, on segmentation and object detection tasks, particularly concerning the precision (P) and recall (R) metrics for classes bounding boxes (B) and mask (M), as well as the mAP50. The various configurations, such as nano (n), small (s), medium (m), large (l), and x-large (x), show that the models preserve their robustness and adaptability, whatever the level of complexity.

The YOLO models evaluated must both accurately detect the position of quadrants in the form of B, with detection performance for this class evaluated using the metrics P(B), R(B), mAP50(B), and mAP50-95(B), but also accurately segment the contour of quadrants in the form of M. Similarly, performance for this second class is

Table 1 Comparative analysis for each variant of the YOLOv5-seg (v5-seg), YOLOv7-seg (v7-seg), YOLOv8-seg (v8-seg), and YOLOv9-seg (v9-seg) architectures for quadrant segmentation and detection

Model	P (B)	R (B)	mAP50 (B)	mAP50-95 (B)	P (M)	R (M)	mAP50 (M)	mAP50-95 (M)
v9c-seg	0.993	0.994	0.995	0.767	0.993	0.994	0.995	0.760
v9e-seg	0.998	1.000	0.995	0.767	0.998	1.000	0.995	0.760
v8n-seg	0.998	1.000	0.995	0.739	0.998	1.000	0.995	0.748
v8 s-seg	0.994	0.996	0.995	0.749	0.994	0.996	0.995	0.748
v8 m-seg	0.997	0.997	0.995	0.752	0.997	0.997	0.995	0.760
v8 l-seg	0.997	0.996	0.995	0.743	0.997	0.996	0.995	0.757
v8x-seg	0.996	0.989	0.995	0.741	0.996	0.989	0.995	0.749
v7-seg	0.998	1.000	0.995	0.726	0.999	1.000	0.995	0.743
v5n-seg	0.989	0.998	0.995	0.718	0.989	0.998	0.995	0.731
v5 s-seg	0.997	0.997	0.995	0.716	0.997	0.997	0.995	0.731
v5 m-seg	0.996	0.993	0.995	0.718	0.996	0.993	0.995	0.737
v5 l-seg	0.992	0.993	0.995	0.717	0.992	0.993	0.995	0.732
v5x-seg	0.996	0.993	0.995	0.718	0.996	0.993	0.995	0.737
Hamamci et al. [89]	–	–	0.651	–	–	–	–	–
He et al. [90]	–	–	0.679	–	–	–	–	–

Highlighted the best results for each metric in bold

evaluated using the metrics P(M), R(M), mAP50(M), and mAP50-95(M). The high scores on all metrics for B and M demonstrate the models' ability to robustly solve these two complementary tasks of object detection and segmentation.

P and R values are generally very high (over 0.98), suggesting high accuracy and complete coverage in the detection of instances by the models.

The different model variants showed very high and very similar R scores. However, YOLOv9e-seg, YOLOv8n-seg, and YOLOv7-seg distinguished themselves by obtaining the highest recall score, with R(B) = R(M) = 1.0.

R(B) measures the model's ability to correctly detect the quadrants present. High recall indicates that the model is able to identify the vast majority of quadrants, thus reducing cases of missed quadrants. This is essential for accurate detection of dental quadrants, as any missed quadrants could lead to diagnostic or treatment errors. On the other hand, R(M) recall assesses the model's ability to perform accurate quadrant segmentation. A high recall measure for segmentation indicates that the model succeeds in correctly delineating quadrant contours in the image. This is important for accurate results when measuring quadrant-specific features such as position, shape, or density.

By using both R(B) and R(M), we obtain a comprehensive assessment of the model's performance in detecting and segmenting dental quadrants. A good balance between these two measures guarantees both accurate detection of the quadrants present and accurate segmentation of their contours.

The YOLOv9e-seg, YOLOv8n-seg, and YOLOv7-seg models also achieved the highest precision in terms of quadrant detection and segmentation, with P(B) and P(M) precision values of 0.998, and precisely P(M)=0.999 for YOLOv7-seg. The other variants also achieved high levels of precision.

High precision is essential in quadrant detection and segmentation, as it ensures that the results obtained are highly reliable and accurate. By minimizing F_P , the YOLOv9e-seg, YOLOv8n-seg, and YOLOv7-seg models are able to identify positive cases with high accuracy, which is crucial for applications such as dental quadrant detection and segmentation.

It is interesting to note that the different variants of the YOLOv5-seg, YOLOv7-seg, YOLOv8-seg, and YOLOv9-seg models all achieved excellent performance in terms of dental quadrant segmentation and detection.

Indeed, these models achieved a score of $\text{mAP50 (B)} = \text{mAP50 (M)} = 0.995$, indicating very high accuracy in locating quadrants in radiographic images.

This high value of mAP50 , at a recognition threshold of 0.5 for both the segmentation (B) and detection (M) tasks, testifies to the remarkable ability of these models to detect and segment quadrants with extreme reliability. This translates into a very low probability of false detections, demonstrating the robustness of these approaches based on YOLO architectures.

It is important to emphasize that this high performance was achieved consistently, regardless of which variant of the YOLO model was used (v5, v7, v8, or v9). This underlines the robustness and versatility of these models in meeting the challenges of automated analysis of dental structures from panoramic X-rays.

In order to compare performance with the state of the art, it is important to note that the methods proposed by Hamamci et al. [89] and He et al. [90] achieved mAP50 s of 0.651 and 0.679, respectively. Although these results are satisfactory, they nevertheless demonstrate the limitations of these approaches in achieving accurate and complete segmentation of the various dental regions.

In contrast, the YOLO-seg models tested in this study have the advantage of directly extracting features at the level of each quadrant without the need for intermediate steps. This capability enables them to achieve significantly better performance than state-of-the-art methods. This result underlines the significant progress made by YOLO architectures to meet the challenges of dental quadrant segmentation.

The mAP50-95(B) scores range from 0.716 to 0.767, while the mAP50-95(M) scores are between 0.731 and 0.76.

The YOLOv9e-seg and YOLOv9c-seg models are distinguished slightly with an mAP50-95(B) score of 0.767, suggesting that they achieve accurate object detection over an IoU interval ranging from 0.5 to 0.95 for bounding box evaluation. Its mAP50-95(M) score of 0.76 also suggests excellent performance in accurate mask segmentation over the same IoU interval of 0.5 to 0.95 for mask evaluation.

The YOLOv7-seg model has slightly lower mAP50-95 scores than YOLOv9(e,c)-seg and YOLOv8(n,s,m,l,x)-seg models, with $\text{mAP50-95(B)}=0.726$ and $\text{mAP50-95(M)}=0.743$. As for the YOLOv5(n,s,m,l,x)-seg models, they generally have the lowest mAP50-95 scores, around 0.716 to 0.718 in terms of detection (B) and around 0.731 to 0.737 in terms of segmentation (M).

These results indicate that the various models succeeded in detecting and segmenting quadrants with overall high accuracy over a range of intervals from 50% to 90%. The mAP50-95 scores obtained with the different models also show good detection and segmentation performance. Nevertheless, YOLOv9e-seg achieved the highest performance, with a slight advantage in terms of mAP over this IoU interval, for both boxes and masks.

In Fig. 5, the original image 5a corresponds to a conventional grayscale panoramic dental X-ray, in which the teeth and bone structures around the jaw are visible.

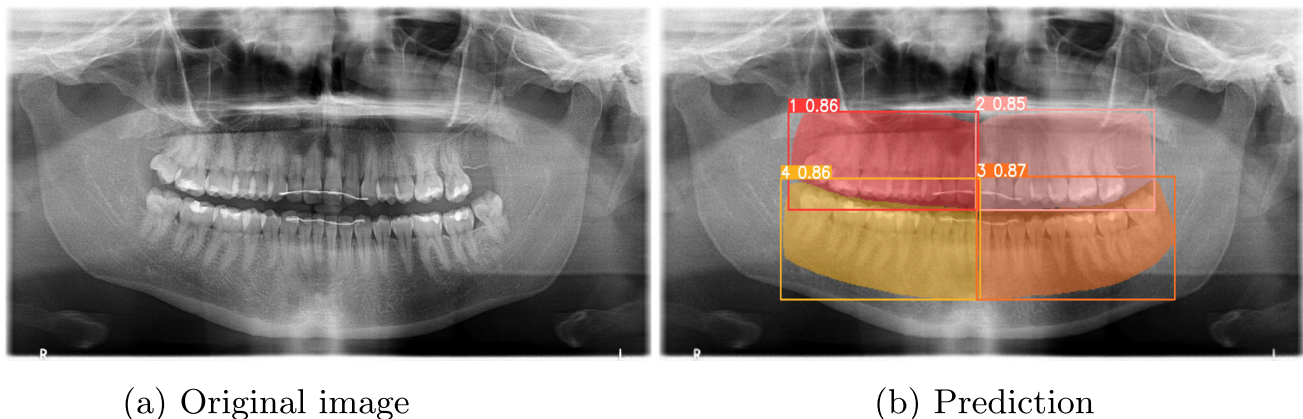


Fig. 5 Comparison of original image (a) and YOLOv9e-seg model prediction (b) for quadrant segmentation.

The predictive image 5b from the YOLOv9e-Seg model also shows the contours of the four dental quadrants, which have been precisely segmented. Each quadrant is delineated by a colored line, allowing it to be visually isolated.

This automated quadrant segmentation is an added value over the original image, as it facilitates the individual analysis of each maxillary and mandibular zone. Here, deep learning models are able to interpret the X-ray image and extract useful anatomical information for further treatment.

The predictive image 5b demonstrates the ability of the YOLOv9e-Seg model to autonomously segment the different quadrants present on a dental panoramic X-ray. This step represents an essential first step in a complete automated analysis aimed ultimately at pathology detection.

4.2 Tooth numbering

Tooth numbering enables identification for diagnostic purposes. The oral cavity is divided into 4 quadrants, each containing 8 teeth numbered from 1 to 8. In each quadrant, teeth 1 and 2 are the incisors, tooth 3 is the canine or cusp, teeth 4 and 5 are the premolars, and teeth 6, 7, and 8 are the molars, of which tooth 7 is the first permanent molar at age 12 and tooth 8 is the wisdom tooth. The numbering system combines the quadrant code and the number of the tooth within it, so tooth 14 is the upper right premolar, tooth 21 is the first upper left incisor and tooth 48 is the lower right wisdom tooth. Since there are 4 quadrants of 8 teeth, an adult has a total of 32 permanent teeth, and this system enables each tooth to be identified in a standardized way for diagnosis or description. To this end, our dataset, specifically dedicated to Tooth enumeration, is divided into three subsets: training, validation, and testing.

Table 2 compares the performance of different variants of the YOLO models, such as YOLOv5-seg, YOLOv7-seg, YOLOv8-seg, and YOLOv9-seg, optimized for the task of segmentation and the individual numbering of the teeth in each oral quadrant, evaluating both the detection of the bounding boxes (B) and the segmentation of the masks (M) corresponding to the teeth and the quadrants.

The models considered (YOLOv5-seg, YOLOv7-seg, YOLOv8-seg, and YOLOv9-seg) are compared according to their ability to distinguish and isolate each tooth in the four quadrants, as well as to correctly assign a

Table 2 Comparative analysis for each variant of the YOLOv5-seg (v5-seg), YOLOv7-seg (v7-seg), YOLOv8-seg (v8-seg), and YOLOv9-seg (v9-seg) architectures for tooth enumeration

Model	P (B)	R (B)	mAP50 (B)	mAP50-95 (B)	P (M)	R (M)	mAP50 (M)	mAP50-95 (M)
v9c-seg	0.937	0.958	0.969	0.603	0.923	0.944	0.948	0.537
v9e-seg	0.968	0.962	0.978	0.608	0.953	0.947	0.959	0.540
v8n-seg	0.918	0.939	0.961	0.591	0.896	0.916	0.928	0.522
v8 s-seg	0.921	0.910	0.956	0.580	0.896	0.891	0.922	0.497
v8 m-seg	0.942	0.925	0.967	0.597	0.920	0.904	0.934	0.518
v8 l-seg	0.947	0.949	0.964	0.597	0.922	0.926	0.93	0.521
v8x-seg	0.944	0.941	0.959	0.593	0.924	0.922	0.930	0.520
v7-seg	0.825	0.878	0.894	0.536	0.812	0.864	0.874	0.464
v5n-seg	0.758	0.886	0.868	0.516	0.747	0.861	0.837	0.454
v5 s-seg	0.423	0.833	0.627	0.359	0.416	0.807	0.605	0.319
v5 m-seg	0.768	0.882	0.873	0.519	0.755	0.859	0.840	0.455
v5 l-seg	0.785	0.875	0.879	0.526	0.770	0.854	0.847	0.456
v5x-seg	0.785	0.875	0.879	0.526	0.770	0.854	0.847	0.456
Hamamci et al. [89]	–	–	0.494	–	–	–	–	–
He et al. [90]	–	–	0.511	–	–	–	–	–

Highlighted the best results for each metric in bold

number to it following the standard tooth numbering system. The analysis also covered not only the precise segmentation of each tooth in its respective quadrant but also the identification of the number assigned to it. The aim is to evaluate the performance of the different models for this key task of automatic indexing of teeth on a dental X-ray. To this end, a dataset has been built up from panoramic X-rays annotated by dental experts.

The precision (P) is a key metric because it indicates to what extent the models manage to delimit well and correctly identify the elements of interest (teeth and quadrants). High P(B) and P(M) scores, such as those obtained by YOLOv9e-seg at 0.968 and 0.953, respectively, show a great ability of these models to precisely locate dental structures.

The recall (R) completes this analysis by evaluating to what extent the relevant elements are detected by the models. High R(B) and R(M) values, also observed for YOLOv9e-seg at 0.962 and 0.947, indicate that the models manage to capture almost all the teeth and quadrants present in the images.

The mAP50 and mAP50-95 metrics provide a global vision of the quality of detection and segmentation. The mAP50, which reaches 0.978 for detection (B) and 0.959 for segmentation (M) with YOLOv9e-seg, illustrates the excellent accuracy of the model at a high confidence threshold (50%). The mAP50-95, slightly lower but still very high, shows that the performance remains remarkable over a wider threshold range.

The comparative analysis highlights the superiority of the YOLOv9e-seg model over the other variants tested. His precision, recall, and mAP scores are consistently the highest, demonstrating his exceptional ability to perform with great reliability the task of precise enumeration of the teeth and their respective quadrants.

The performance achieved with YOLO models, and YOLOv9e-seg in particular is all the more remarkable when compared with the state-of-the-art approaches proposed by Hamamci et al. [89] and He et al. [90] Indeed, the latter only achieve mAP50 (B) of 0.494 and 0.511 respectively, results well below those of YOLO models.

This comparison highlights the significant advances made by YOLO architectures, and YOLOv9e-seg in particular, for the specific tasks of segmentation and tooth enumeration. The superior performance of these models underlines their high efficiency and relevance for this dental application.

This contrast in results underlines the importance of the progress made with YOLO models in this field. Their ability to outperform the most advanced existing approaches demonstrates their strong potential for the development of more efficient and reliable automated dental analysis solutions.

Figure 6 illustrates the results of segmentation and detection of tooth numbers within each maxillary and mandibular quadrant.

This image 6b illustrates the application of the YOLOv9e-seg model for the precise detection and segmentation of teeth as well as the identification of the number of each tooth and the quadrant to which it belongs on a dental X-ray.

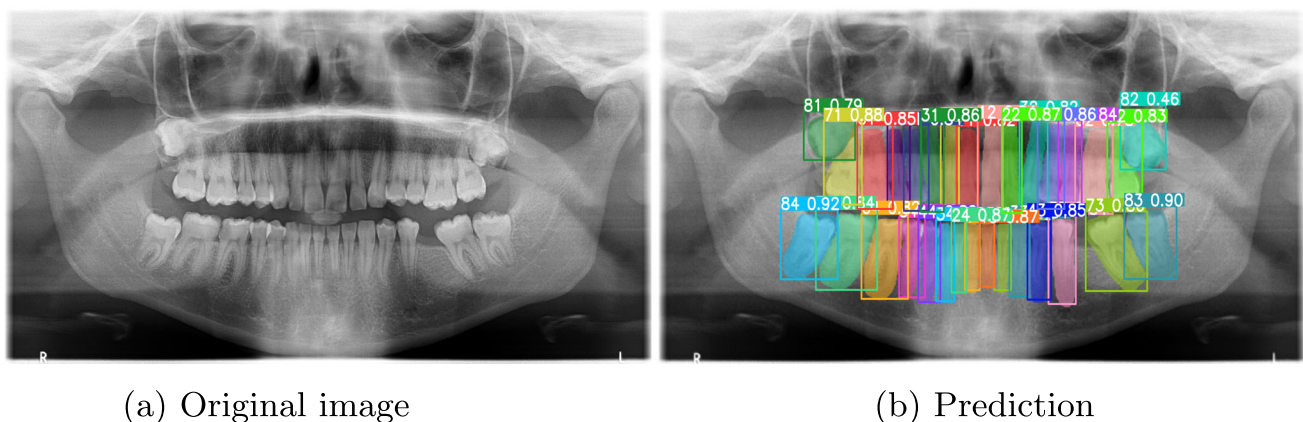


Fig. 6 Comparison of original image (a) and YOLOv9e-seg model prediction (b) for Tooth enumeration.

We can see that each tooth has been individually numbered from 1 to 8 according to its position in the quadrant. Since each quadrant contains 8 teeth, and there are 4 quadrants per jaw, this panoramic X-ray shows a total of 32 numbered teeth.

The boxes delimit each detected tooth and display a number consisting of two digits. The first digit on the left indicates the tooth number, while the second digit corresponds to the number of the quadrant to which the tooth belongs. For example, the box numbered “81” indicates tooth number 8, located in quadrant 1.

This detailed information, provided by the YOLOv9e-seg model, is a very useful tool for medical and dental applications. They allow a precise enumeration and analysis of the tooth by identifying each tooth and locating it in its respective quadrant.

The analysis of these numerical data makes it possible to evaluate the performance of the model for this specific task of enumeration and dental segmentation. The displayed values, such as precision (P), recall (R), and maps, give an indication of the quality and reliability of the model’s predictions.

4.3 Diseases detection

The final stage in the automated diagnostic pipeline is the detection of dental diseases. This follows on from the previous segmentation phases, which delimited the quadrants and digitally identified each tooth.

This step aims to recognize three key pieces of information: the disease affecting the tooth, its number within the dental enumeration system, and the quadrant number to which it belongs.

Performance on this task was not as good as expected. Among the models tested, YOLOv9e proved to be the best performer in pathology recognition, although there was still room for improvement in this key stage of dental disease identification.

In Table 3, it is noted that the YOLOv5n model obtains the best precision score $P(B) = 0.659$, followed by the YOLOv6n and YOLOv9e models (with data preprocessing) which display respective precisions of $P(B) = 0.6$

Table 3 Comparative analysis for each for each variant of the YOLOv5, YOLOv6, YOLOv7, YOLOv8, and YOLOv9 architectures for disease detection

Model	P(B)	R(B)	mAP50 (B)	mAP50-95(B)
v9e (with data preprocessing)	0.530	0.626	0.610	0.447
v9e	0.505	0.423	0.413	0.283
v9c	0.446	0.309	0.287	0.206
v8n	0.467	0.222	0.213	0.149
v8 s	0.399	0.292	0.244	0.171
v8 m	0.453	0.256	0.268	0.187
v8 l	0.423	0.262	0.299	0.223
v8x	0.459	0.305	0.320	0.228
v7	0.407	0.298	0.177	0.130
v6n	0.600	0.125	0.108	0.064
v6 s	0.365	0.31	0.203	0.133
v6 m	0.367	0.252	0.204	0.141
v6 l	0.397	0.301	0.185	0.121
v6x	0.366	0.301	0.213	0.133
v5n	0.659	0.213	0.139	0.090
v5 s	0.512	0.258	0.160	0.107
v5 m	0.515	0.251	0.199	0.143
v5 l	0.389	0.331	0.224	0.157
v5x	0.470	0.325	0.230	0.159
Hamamci et al. [89]	–	–	0.613	–
He et al. [90]	–	–	0.543	–

Highlighted the best results for each metric in bold

and $P(B) = 0.53$. These results demonstrate the capabilities of these models to accurately detect diseased teeth, specifying their number as well as the quadrant to which they belong, thus avoiding false positives during detection.

Despite these remarkable performances of the YOLOv5n and the YOLOv6n, the YOLOv9e model with data preprocessing stands out as the best overall solution for the detection of dental diseases in this comparative table. Indeed, it displays exceptional performances in terms of precision ($P(B) = 0.5$), recall ($R(B) = 0.626$), mAP50 (0.6), and mAP50-95 (0.447).

In comparison, the method of Hamamci et al. [89], based on the detection of points of interest, displays a mAP50 of 0.613, slightly higher than the 0.61 obtained by the YOLOv9e. Similarly, the model proposed by He et al. [90] reaches a mAP50 of 0.543, clearly below the performance of the YOLOv9e.

These improved results of the YOLOv9e were obtained thanks to the effective application of preprocessing techniques and data augmentation. Indeed, we have set up static cropping, extracting a horizontal region between 15% and 83% of the image as well as a vertical region between 30% and 80%, allowing us to focus on the area of interest, while eliminating irrelevant elements. We also used an automatic contrast adjustment method, improving the visibility and distinction of the dental structures on the images.

In addition, we applied various data increments, including 90° rotations to include flipped images, random rotations between -10° and +10° to strengthen the robustness of the model in the face of slight variations in orientation, as well as random changes in brightness, between 0% and +25%, to increase the ability of the model to handle lighting differences.

These pre-treatment techniques and data augmentation have made it possible to significantly improve the performance of the YOLOv9e model, propelling it as the best solution for the precise detection of diseased or infected teeth, with a detailed identification of the tooth number and the corresponding quadrant. This information would be valuable for health professionals, allowing them to quickly target the problem area and develop an appropriate treatment plan. The YOLOv9e is therefore positioned as a particularly promising tool for the precise detection of diseased or infected teeth.

However, further improvements are needed to achieve more robust detection of dental pathologies. Although this methodology has demonstrated its capabilities through the precise localization of teeth and the identification of certain conditions, the reliability of disease recognition could be strengthened.

Optimization of the model during the detection stage should help limit errors in automated diagnosis. Continued development of our image analysis pipeline is essential if we are to achieve optimum performance in the automated detection of oral problems.

Figure 7 shows the results of a dental disease detection prediction performed by the YOLOv9e model.

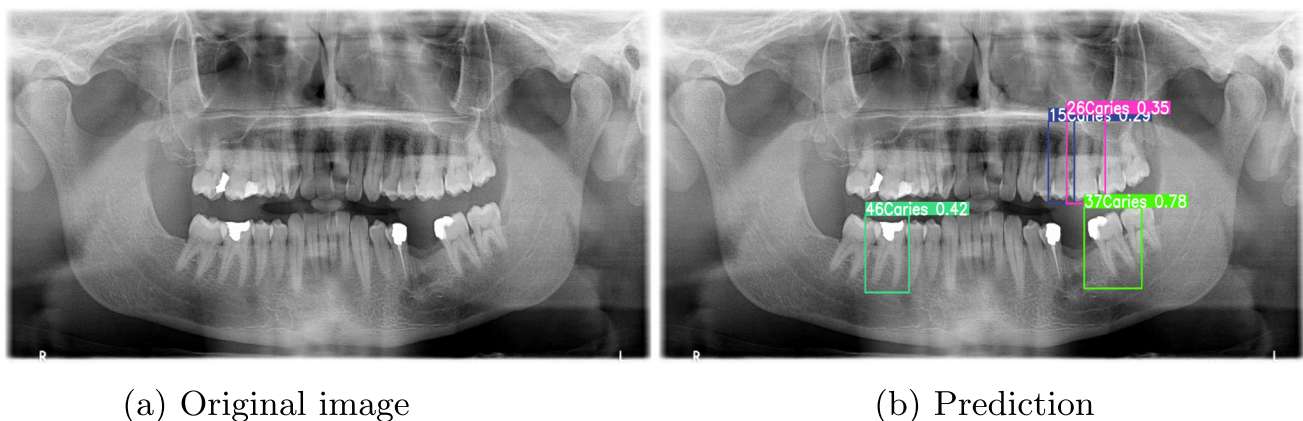


Fig. 7 Comparison of the original image (a) and YOLOv9e model prediction (b) for the Detection of three pieces of information: tooth disease, tooth number, and quadrant number associating it.

Identification of dental quadrants is ensured by their clear numbering, which localizes each anatomical zone of the oral cavity in a standardized way. Within the quadrants, each tooth is individually numbered, enabling precise localization. Certain pathologies also seem to be detected, in the form of radiolucent lesions visible on several teeth, probably reflecting caries or infection.

The image 7b shows a dental panoramic X-ray on which information has been added by a YOLOv9e model for automated tooth detection and analysis. The model has detected 3 teeth with caries, indicated by the information “26Caries” and “37Caries”. Each detected tooth is assigned a number, from 1 to 32, in Line with standard dental numbering, and teeth with numbers 6 and 7 are also associated with their respective quadrant, indicated by numbers 2 and 3. In this case, the model has correctly detected the pathology.

However, tooth “15Caries” was incorrectly located in quadrant 2 instead of quadrant 1, revealing a limitation of the model at the prediction stage.

This prediction represents the culmination of our ML pipeline, although some improvements remain to be made for more reliable detection of dental disease.

This limitation highlights the fact that, although the model demonstrated great robustness for quadrant segmentation, tooth number detection and dental pathology identification, additional efforts are still required to achieve better performance.

Despite this one-off weakness, the overall results obtained by the YOLOv9e model demonstrate its strong potential for the automation of large-scale dental analysis, offering encouraging prospects for the development of ever more effective diagnostic aid solutions.

4.4 Challenges and limitations

Although the YOLOv9e model has stood out as the best solution for the detection and segmentation of dental quadrants, as well as for the precise enumeration of teeth, as shown respectively in comparative tables 1 and 2, with exceptional performance in terms of accuracy, recall and mAP, its use for the detection of dental diseases by identifying the tooth number and the quadrant involved poses several important limitations.

Generic object detection models, such as YOLO, face specific challenges when applied to the dental field. Their architecture, originally designed for general tasks, does not account for the anatomical and pathological specificities of the oral cavity. This inadequacy results in suboptimal performance in the detection of dental structures.

The variable quality of clinical images presents a significant first obstacle. X-rays and intraoral photographs often exhibit variations in contrast, brightness, and resolution. These fluctuations arise from differences in acquisition equipment and the specific conditions of intraoral shooting. Artifacts and visual noise further complicate automatic analysis.

The complexity of dental pathologies represents another important limitation. Conditions such as early caries or initial periodontal lesions exhibit subtle and poorly contrasted visual signs. Distinguishing them requires sharp clinical expertise that standard models do not inherently possess. Moreover, some lesions share similar visual characteristics, increasing the risk of confusion.

The size and diversity of the datasets used to train these models are often insufficient, affecting their learning capacity, performance, and ability to generalize to different cases. For example, a dataset comprising only a few hundred dental images, instead of the thousands required, limits the model’s ability to reliably detect complex patterns. In addition, the lack of diversity in terms of pathologies (such as caries, periodontal disease, and fractures), anatomical variations (tooth shapes or positions as a function of age or genetics), and imaging conditions (including sensor quality, angle, and type) further limits the model’s adaptability to the various situations encountered in practice.

The issue of unbalanced datasets particularly affects this medical domain. The available training sets often contain an overrepresentation of typical cases at the expense of rare or complex clinical situations. This bias limits the models’ ability to recognize unusual anatomical variations or infrequent pathologies.

Individual anatomical peculiarities add an additional layer of difficulty. Dental morphology varies significantly based on the patient's age, therapeutic history, or genetic factors. Malpositioned and impacted teeth present complex spatial configurations that standard algorithms struggle to interpret correctly.

Ultimately, these models cannot accurately identify which specific tooth is affected by a given disease. To overcome these limitations and make YOLOv9e a reliable tool in dentistry, several improvement strategies are essential. Firstly, it is crucial to enrich the datasets with diverse, well-annotated clinical cases. In addition, the adaptation of neural architectures should involve the integration of specific dental knowledge. Finally, close collaboration between artificial intelligence experts and clinicians will facilitate the development of tools that truly meet the practical needs of dental diagnostics.

5 Conclusion

In conclusion, this work has enabled the development of a three-stage pipeline based on machine learning for the automated detection of dental disease from panoramic X-rays. The study demonstrated the effectiveness of the YOLOv8-seg and YOLOv9 models in performing three key tasks: segmentation of the dental quadrants, detection of individual teeth, and identification of present pathologies.

The results highlight the potential benefits of integrating AI into dental X-ray analysis. This approach enables faster, more accurate image analysis while reducing diagnostic errors and promoting early detection of dental problems. In addition, grouping these different tasks into a single model offers more efficient and useful results for diagnostic support.

However, the model has also shown certain limitations, particularly in the reliable detection of more subtle pathologies. It will therefore be important to continue training the model on larger and more diverse datasets in order to enhance its detection capabilities, particularly for more complex cases. In parallel, it would be interesting to explore complementary approaches based on the classification of dental pathologies. These methods, combined with the segmentation and detection performances already demonstrated, could provide an even more complete and precise understanding of the various diseases present.

The integration of classification models specialized in the identification of caries, infections, alignment problems, or other pathologies could indeed fill the gaps observed in the most delicate cases. This synergy between segmentation, detection, and classification would provide dental health professionals with an even more powerful and reliable analysis tool, greatly facilitating diagnosis and patient follow-up.

The continuous improvement of the model, through the enrichment of training data and the addition of targeted classification approaches, will enable the development of a more powerful and versatile solution for the automated analysis of dental X-rays. This will open up new prospects for more effective and earlier management of oral health problems.

Finally, the addition of an explainability approach will also be necessary to enable a better understanding of the model's decisions by clinicians and thus foster its acceptance and trust.

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Data availability This work uses a publicly available dataset; see reference [66] for data availability. More details about this dataset are available in Sect. 3.1.

Declarations

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Conflict of interest The authors declare no Conflict of interest.

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